SeeLess: Pitch Document

Nov 13, 2026 @Sanskar , @Gargi

Solution pitch

Every scene, heard. SeeLess adds audio description to any MP4 so blind and low-vision viewers can follow what is on screen. Upload a video; get back a version with spoken narration placed only in the silent gaps between dialogue.

- **Hear:** Gnani AI Speech-to-Text marks exactly when each line of dialogue starts and ends.
- **See:** Gemini describes actions, expressions and scene continuity. One part of the pipeline, not the core.
- **Speak:** Gnani AI TTS Timbre v2.5 voices each description, dropped into a gap with zero overlap.

Built on Gnani AI STT + TTS Timbre v2.5 · Gemini · FastAPI · Celery + Redis · FFmpeg · Next.js 15

Repo: github.com/2823sanskar/SeeLess

Built for the #GreatIndianAIInternshipChallenge by Gnani AI.

Problem

Most films and videos ship without audio description. Blind and low-vision viewers hear the dialogue but miss the actions, expressions and scene changes that carry the story.

Manual description is slow and costly. Naive AI narration talks over dialogue, which makes the result worse than silence.

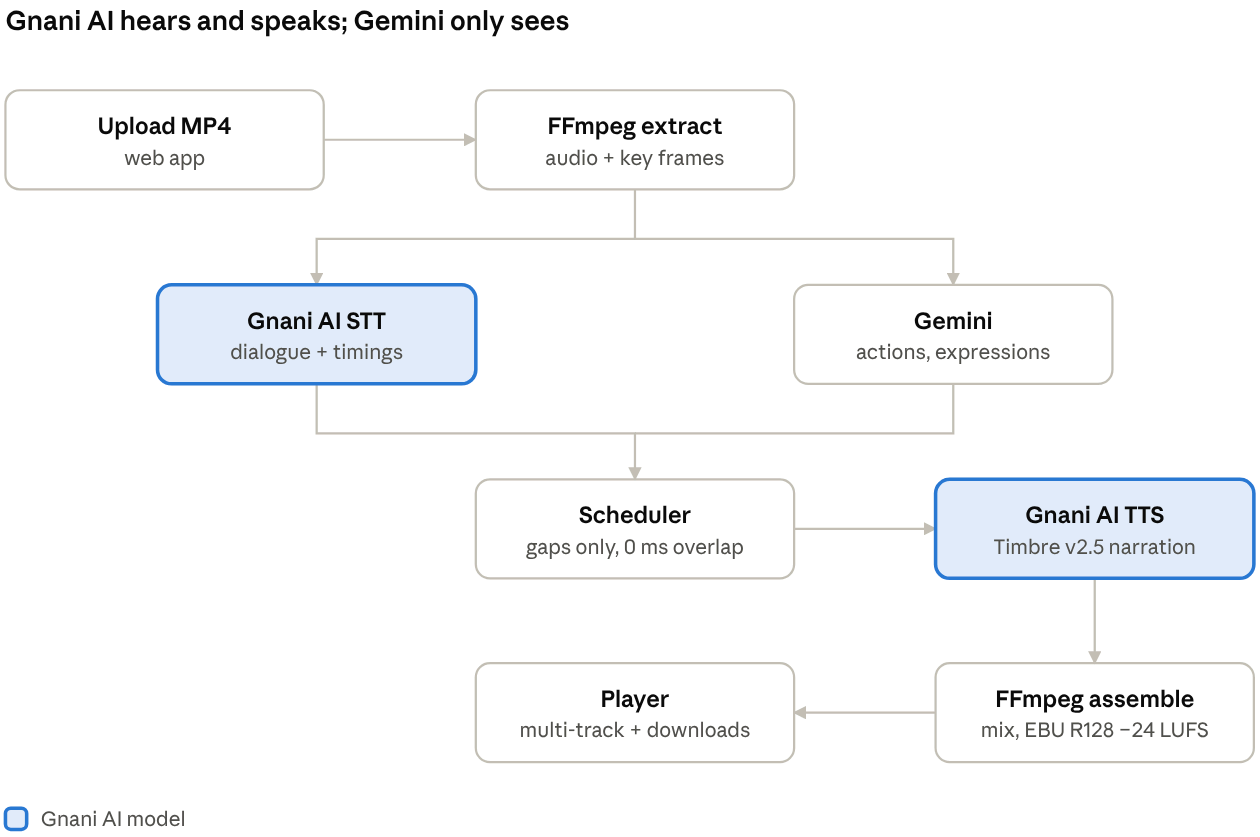
What SeeLess does

- Upload an MP4; audio and key frames are extracted.
- Dialogue-only transcript with speech intervals and safety padding.
- Visual descriptions of actions, expressions and scene continuity.
- Deterministic scheduler fits narration into silent gaps: 0 ms overlap.
- Natural narration from Gnani AI Timbre v2.5.

- Accessible multi-track player: switch original and described audio in one tap.
- Downloads: described MP4, WAV, transcript JSON.
- EBU R128 loudness at −24 LUFS, with ducking under dialogue.

How it works

One upload runs through a background pipeline. Gnani AI handles both audio ends.



SeeLess pipeline · 8 steps, 2 Gnani AI models

Extraction fans out to Gnani AI STT and Gemini; the scheduler merges both, then Gnani AI TTS voices the result.

1. **Upload:** MP4 in; FFmpeg/FFprobe extract audio; PySceneDetect and OpenCV pick key frames.
2. **Transcribe:** Gnani AI STT returns dialogue only, with speech intervals and safety padding.
3. **Describe:** Gemini turns key frames into descriptions of actions, expressions and scene continuity.

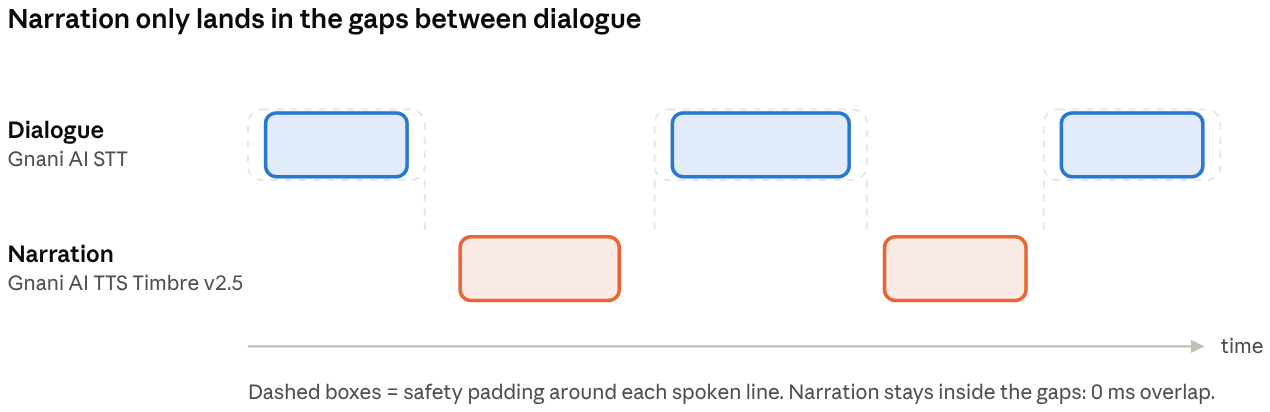
4. **Schedule:** a deterministic scheduler places each description in a silent gap. Zero overlap with dialogue.
5. **Speak:** Gnani AI TTS Timbre v2.5 voices each description.
6. **Assemble:** FFmpeg and PyDub mix narration into the video, normalised to EBU R128 −24 LUFS with ducking.
7. **Play and download:** multi-track player; MP4, WAV and transcript JSON exports.

Gnani AI models

Two Gnani AI models carry every sound in SeeLess.

Model	Role	What it gives SeeLess
Gnani AI Speech-to-Text	Hears	Dialogue-only transcript, speech interval boundaries, safety padding
Gnani AI TTS Timbre v2.5 (`timbre-v2.5`)	Speaks	Natural, expressive narration for each description, exported as WAV

Gemini supplies visual context only. It never decides timing or voice.



Audio timeline · dialogue vs narration

STT boundaries plus padding define the gaps; TTS narration is placed only inside them.

Technical design and key decisions

Stack

Part	Tech
Frontend	Next.js 15, React 19, TypeScript, Tailwind
Backend	FastAPI, SQLAlchemy, SQLite
Jobs	Celery, Redis
Speech	Gnani AI STT, Gnani AI TTS Timbre v2.5
Vision	Gemini
Media	FFmpeg/FFprobe, PySceneDetect, OpenCV, PyDub

Key decisions

We chose	Instead of	Win
Gnani AI STT for dialogue only	Describing over a full transcript	Exact speech boundaries to schedule around
Safety padding around every spoken line	Cutting at raw timestamps	Narration never clips a word
Deterministic scheduler	Asking an LLM to place narration	Zero overlap, repeatable output
Gnani AI Timbre v2.5 for narration	Generic synthetic voice	Natural, expressive description
Gemini for visuals only	One model for everything	Gnani AI owns hearing and speaking; vision stays swappable
Celery + Redis background jobs	Request-time processing	Long videos don't block the API
EBU R128 at −24 LUFS with ducking	Raw mix levels	Even loudness; dialogue stays clear

Demo and links

- **GitHub:** github.com/2823sanskar/SeeLess
- **Demo video**:
- **Demo flow:** Upload → Describe → Play
- **Output:** described MP4, WAV, transcript JSON

Team

Team: Sanskar Ubale, Gargi Garad.

SeeLess is ready to demo: upload any MP4, get described audio, play it in the accessible player.